Lecture 3: Regularization and Model Stability

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Why Stability Matters in Forecasting

Definition: Coefficient stability

How consistently estimated coefficients keep their sign and magnitude across nearby training windows.

Definition: Bias–variance trade-off

Accepting a small increase in bias to obtain a larger reduction in variance and more reliable out-of-sample forecasts.

Teaching reminder. In this lecture, *stability* and *forecast accuracy* are evaluated together; neither metric alone is sufficient for model choice.

Key idea. The same target and predictor set can produce very different coefficients when the training window changes.

Forecasting context.

- Scenario: predict next-day S&P 500 return using lagged returns of MSFT, AAPL, and VIX.
- In January, model coefficients look reasonable; in March, refitting on a new window changes signs and magnitudes.
- Message to students: this is not a coding bug, it is a **forecasting risk signal**.
- Distinguish **accuracy** (RMSE) from **reliability** (coefficient stability and interpretability over time).

Concept contrast.

Metric	What it measures	Main failure mode
In-sample R^2	Description of historical data	Overfitting / noise memorization
Out-of-sample RMSE	Forecast performance on unseen data	Regime change / weak signal
Coefficient stability	Structural consistency of model logic	Multicollinearity / sample sensitivity

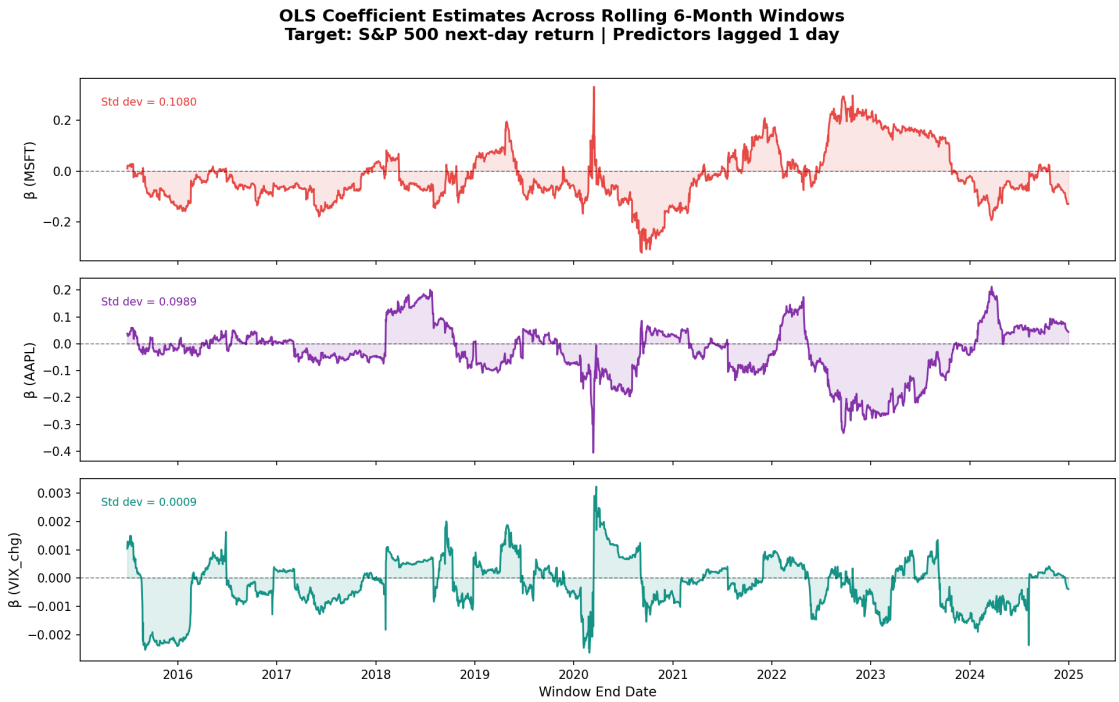

Figure 1: Coefficient instability across rolling windows: the same OLS specification produces different stories when the sample window changes.

Implication. If OLS is sensitive to small window shifts, we need a method that constrains coefficient behavior.

Multicollinearity and Coefficient Volatility

Definition: Multicollinearity

A predictor structure where explanatory variables are strongly correlated, making OLS attribution fragile.

Core intuition. When predictors carry overlapping information, OLS struggles to allocate contribution cleanly.

- Analogy: if two siblings always submit group work together, it is difficult to assign separate credit.

- In forecasting data, highly correlated predictors create the same attribution problem.
- Small noise in one sample can produce large coefficient changes in another sample.

Technical statement.

$$\hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y$$

When predictors are highly correlated, $X^\top X$ becomes close to singular. The inversion amplifies noise, making $\hat{\beta}_{\text{OLS}}$ unstable.

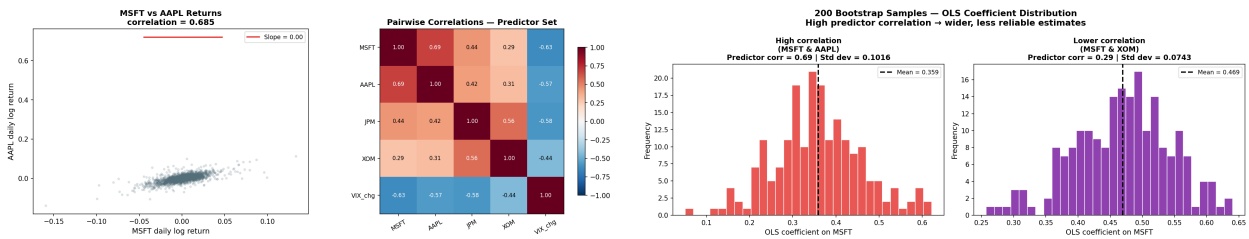

Figure 2: Left: overlapping predictors create a multicollinearity geometry problem. Right: OLS coefficients become highly volatile across samples.

Forecasting implication.

- Volatile coefficients generate volatile forecasts.
- A model can look acceptable in-sample but still be unreliable in production use.
- We need an intentional constraint on coefficient size.

Ridge Regression: Stability-Oriented Shrinkage

Definition: Regularization

Adding a penalty term to the loss function to constrain coefficient size and reduce variance.

Definition: Ridge regression (L_2)

Regularization using $\sum_{j=1}^{p} \beta_j^2$, which shrinks all coefficients smoothly but usually does not set them to zero.

Core idea. Add an L_2 penalty so coefficients are shrunk smoothly toward zero.

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$$

Adding λI stabilizes the inversion and directly addresses multicollinearity-induced variance.

- $\lambda = 0$: ridge equals OLS.
- Larger λ: stronger shrinkage, lower variance, more bias.
- Business interpretation: a slightly biased but stable model is often preferred for institutional forecasting.
- Hyperparameter tuning must be done using validation RMSE, not in-sample fit.

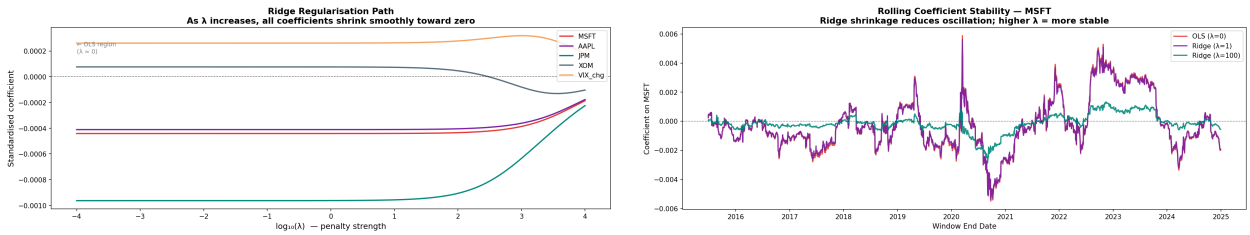

Figure 3: Ridge coefficient path and stability gain: coefficients shrink smoothly as λ increases, improving robustness across samples.

Key takeaway. Ridge does not delete variables; it keeps all predictors and controls their magnitudes.

Lasso Regression: Shrinkage and Sparsity

Definition: Lasso regression (L_1)

Regularization using $\sum_{j=1}^{p} |\beta_j|$, which can shrink some coefficients exactly to zero.

Core idea. Replace L_2 with L_1 penalty to combine shrinkage and feature selection.

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

Because of the L_1 geometry, lasso can set some coefficients exactly to zero.

- Practical value: automatic variable screening when there are many candidate features.
- Communication value: sparse models are easier to explain to non-technical stakeholders.
- Caution: in finance, many signals are weak; lasso may discard useful but small effects.
- Stability caution: selected variables may flip across windows.

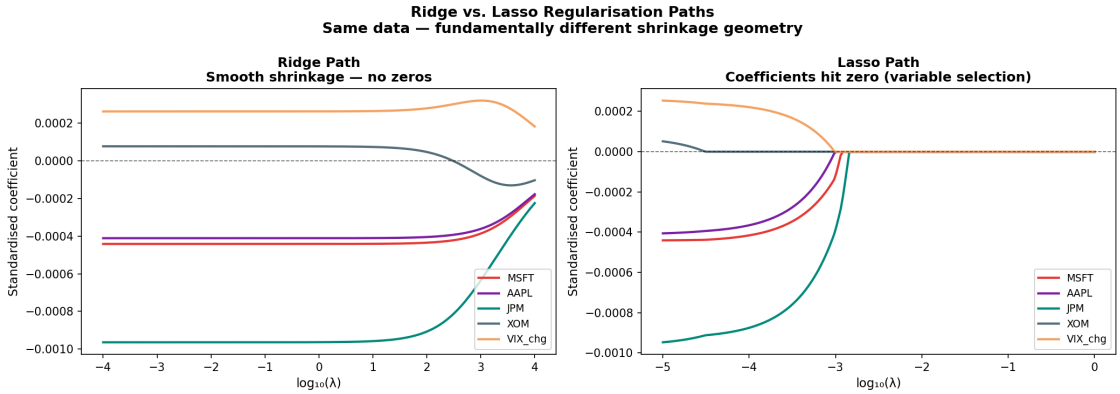

Figure 4: Lasso vs. ridge coefficient paths: lasso creates exact zeros (selection), while ridge keeps all variables with smooth shrinkage.

Ridge vs. Lasso: Forecasting Decision Rules

Decision principle. Choose according to forecasting objective, not algorithm popularity.

	Ridge	**Lasso**
Penalty	$L_2 : \sum \beta_j^2$	$L_1 : \sum \lvert\beta_j\rvert$
Shrinkage style	Smooth and continuous	Sharp, can hit zero
Variable selection	No	Yes
Best use case	Many correlated predictors, all likely relevant	High-dimensional setting with sparse true signal
Stability across windows	Usually higher	Can vary due to selection flips

- If governance, robustness, and stable behavior matter most, start with ridge.
- If interpretability via feature screening is primary, evaluate lasso.
- In weak-signal financial returns, ridge is often more reliable over repeated re-estimation.

Implementation Workflow: OLS, Ridge, and Lasso

Experimental protocol.

1. Training set: first 70% of observations for model fitting.
2. Validation set: next 15% for selecting λ (ridge and lasso).
3. Test set: final 15% used exactly once for final comparison.

Modeling sequence.

- Build one OLS baseline, one ridge model, and one lasso model on identical features.

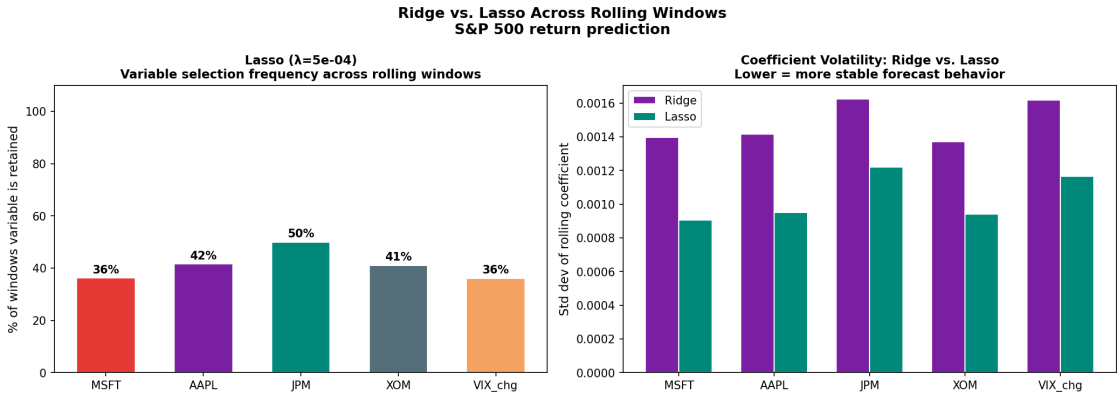

Figure 5: Practical comparison of ridge and lasso under forecasting objectives: stability-oriented vs. sparsity-oriented modeling choices.

- Tune hyperparameters only on validation data.
- Compare test RMSE and rolling-window coefficient stability.
- Write a short recommendation memo in manager-facing language.

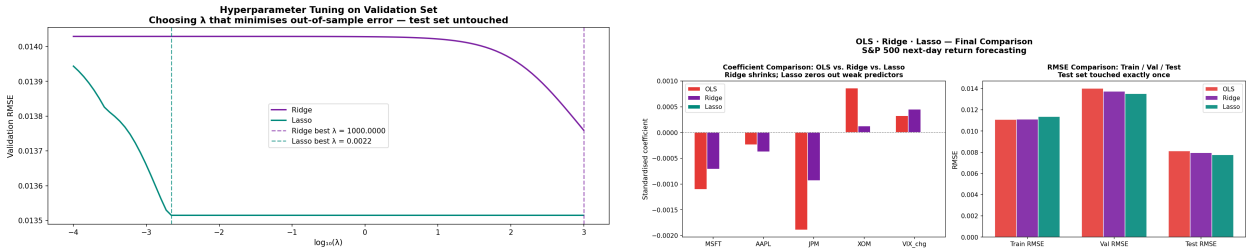

Figure 6: Hyperparameter tuning and final model comparison workflow used in class for Assignment 2 preparation.

Model recommendation memo template.

- **Task:** Forecast next-day S&P 500 log return using lagged returns (MSFT, AAPL, JPM, XOM) and VIX change.
- **Evidence:** OLS, ridge, and lasso often have similar test RMSE in weak-signal daily returns.
- **Recommendation:** Prefer ridge when stability and institutional interpretability are the priority.
- **Exception:** Prefer lasso when feature universe is large and explicit variable screening is required.